

CS 103000

Prof. Madeline Blount

Week 4:

LOOPS (part 1)



<https://airtable.com/appLDEvuelyDRol2b/shrjBerO7c6dINYQy>

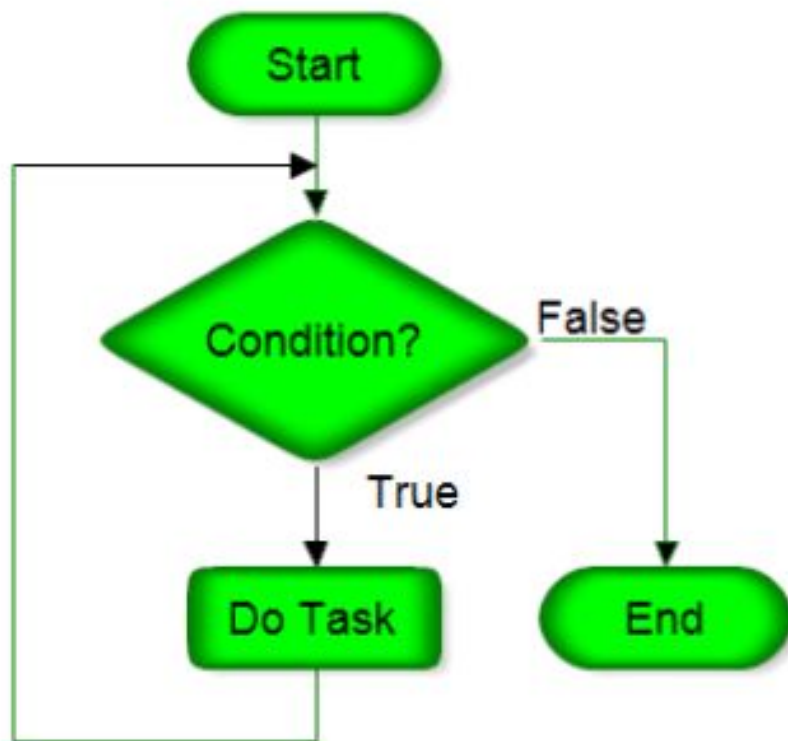


Dall-E 2: cats learning C++ in the forest on '90's technology



<https://airtable.com/appLDEvuelyDRoI2b/shrjBerO7c6dINYQy>

While Loop





```
while (raining)
```

what is RANDOM?

- Elusive - easier to describe what it *isn't*
- No discernable **pattern**
- Not **predictable**
(Reveals itself in **sequence**)
- Not **deterministic** (the more factors we know, the more causality or pattern we can find, then it's not random)

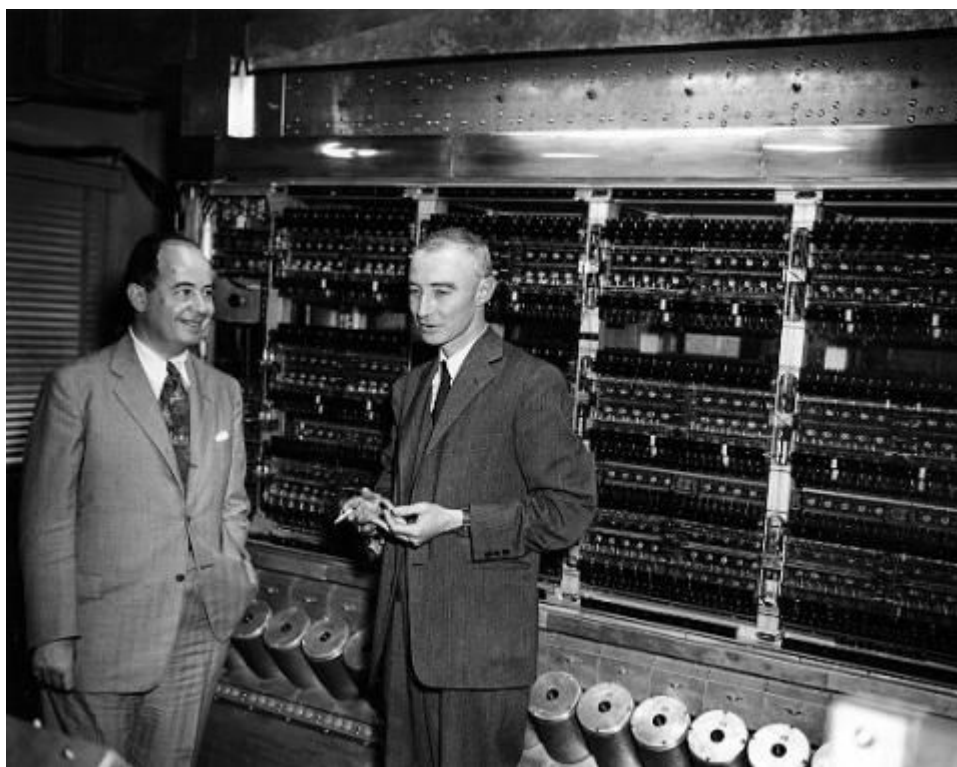
BUT ...

computers ARE deterministic

SAME INPUTS -> SAME OUTPUTS



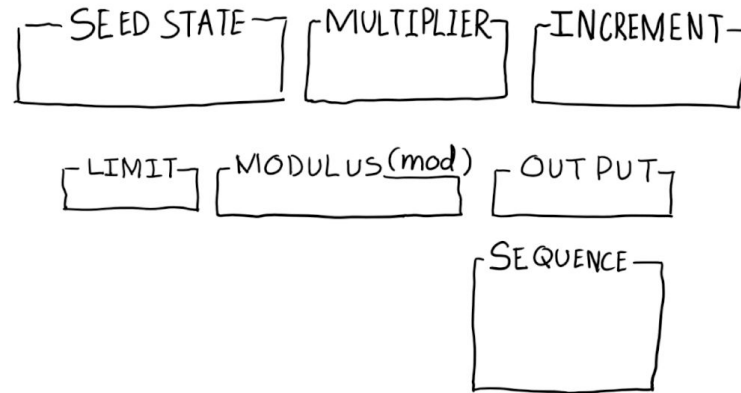
SNL



John von Neumann (here w/Robert Oppenheimer):
"Anyone who considers arithmetical methods of
producing random digits is, of course, in a
state of sin."

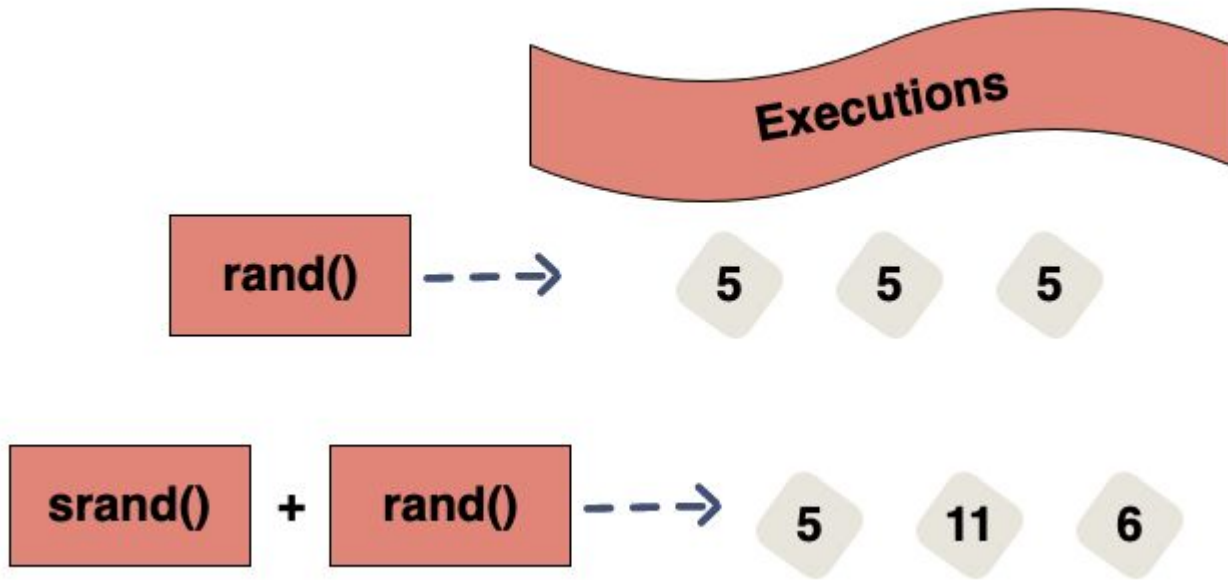
Pseudorandom! a **simulation**, random enough

- `rand()` = Linear Congruential Generator (LCG)
- $x = ((a * x) + c) \% m$
- Next number based on the previous (**state**)





`srand()` = SEED



RANDOMNESS FROM SEED ONLY; SAME SEED = SAME SEQUENCE!

Random integer between 0 and 1:

```
rand() % 2;
```

Random integer between 0 and 9:

```
rand() % 10;
```

Random integer between 0 and 10:

```
rand() % 11;
```

Random integer between 10 and 20:

```
rand() % 11 + 10;
```